Section A (40 Marks)

1 **Workout: 7042 + 2513**

$$\begin{array}{r} 7042 \\ +\,2513 \\ \hline 9555 \end{array}$$

2 **Write 71090 in words.**

Thousands	units
71	090

Seventy one thousand ninety.

3 **Find the sum of all triangular numbers less than 12.**

= 1, 3, 6, 10

Sum = 1+3+6+10

= 20

4. **Find the number of elements in a set with 16 subsets.**

No. of C = 2^n

$16 = 2^n$

$2^4 = 2^n$

$4 = n$

$\therefore n = 4$ elements

$16 = 2^4$

5 **Write 0.00503 in standard form.**

$0.00503 \times 10 = 0.0503$

$0.0503 \times 10 = 0.503$

$0.503 \times 10 = 5.03$

$= 5.03 \times 10^{-3}$

6 **A school concert started at 9:30am and ended at 4:00pm. How long did the concert last?**

ST = +9:30 00:00 = 0930hrs

ET = +4:00 12:00 = 1600hrs

Duration = ET − ST

Duration = 1600 − 0930

= 630

= 6½ hours

7 **Using a ruler and a pair of compasses, construct an angle of 135°**

135°

8 **Given that n = 3, m = -2 and y = 8, find the value of $\frac{mn}{y+m}$**

$= \frac{mn}{y+m}$

$= \frac{-2 \times 3}{8+(2)}$

$= \frac{-6}{8-2}$

$= \frac{-6}{6}$

$= -1$

9 **Change 4.56 tonnes to kilogrammes**

1 tonne = 1000kg

4.56 = $\left(\frac{456}{100} \times 1000\right)$ kg

= 4560 tonnes.

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10 In a box there are blue and red pens. The probability of picking a blue pen from the box at random is $\frac{5}{7}$. If there are **10** red pens in the box, find the total number of pens in the box.

$P(B) = \frac{5}{7}$

$P(R) = \frac{7}{7} - \frac{5}{7}$

$= \frac{2}{7}$

Let the total no be x.

$\frac{2}{7} \times x = 10$

$\frac{2}{7}x = 10$

$7 \times \frac{2}{7}x = 10 \times 7$

$\frac{2x}{7} = \frac{5}{10} \times 7$

$x = 35$ pens

11 If today is Saturday. What day of the week will it be **50** days from now?

$6 + 50 = -(\text{finite } 7)$

$56 = -(\text{finite } 7)$

$\frac{56}{7} = 8 \text{ rem } 0$

$= 0 (\text{finite } 7)$

∴ The day will be Sunday.

12 John and Peter were told to report to the police station after **15** days and **16** days respectively. After how many days will they report to the police station together again?

2	15	16
2	15	8
2	15	4
2	15	2
3	15	1
5	5	1
	1	1

$= 2 \times 2 \times 2 \times 2 \times 3 \times 5$

$= 240$ days

13 Work out: **18.43 - 9.16 + 7.23**

$= 18.43 + 7.23 - 9.16$

$= 25.66 - 9.16$

$25.66 - 9.16 = 16.50$

$= 16.5$

$18.43 + 7.23 = 25.66$

14 Hadijjah bought $1\frac{1}{2}$ kg of rice at **sh.1800** for every a half kg. Find how much Peter spent altogether.

$\frac{1}{2}$ kg $=$ sh. 1800

$1\frac{1}{2}$ kg $= \left(\frac{3}{2} \times 1800\right) \div \frac{1}{2}$

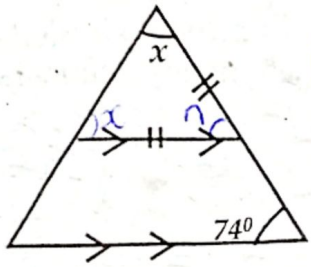
$= \left(\frac{3}{7_1} \times \frac{1800}{1}\right) \div \frac{1}{2}$

$= 2700 \div \frac{1}{2}$

$= 2700 \times \frac{2}{1}$

$= $ Sh. 5400.

15 **Study the figure below and find the value of x in degrees.**



Find x

$\angle n = 74°$ (corresponding)

$x + x + 74° = 180°$

$2x + 74° = 180°$

$2x + 74° - 74° = 180° - 74°$

0

$\frac{1}{2_1}x = \frac{53}{106}$

$x = 53°$

16. A motorist covers **150km in 2 hours.** How long will he take to cover a distance of 450km?

150Km take 2hours

450Km take $\left(\frac{450 \times 2}{150}\right)$ hrs

$= 6$ hours

17. Solve $\frac{2}{3}P^2 = 24$

$\frac{1}{3} \times \frac{2}{3} P^2 = 24 \times 3$

$\frac{1}{1} \frac{2}{1} p^2 = \frac{12}{24} \times 3$

$p^2 = 36$

$\sqrt{p^2} = \sqrt{36}$

$p = 6.$

18. Workout the mean of **-4, 0, 9, 4 and 6**

mean $= \frac{\text{sum of items}}{\text{no of items}}$

$= \frac{9+6+4+0+(-4)}{5}$

$= \frac{19-4}{5}$

$= \frac{15}{5}$

$= 3$

19. The circumference of a circle is **132dm.** Calculate its radius.

$C = 2\pi r$

$132 = 2 \times \frac{22}{7} \times r$

$7 \times 132 = 44r \times 7$

$7 \times 132 = 44r$

$\frac{44r}{44} = \frac{7 \times 132}{44}$

$r = 7 \times 3$

$r = 21$ dm

20. Omoding covered a distance of **60km** in **1** hour and **15 minutes.** Calculate his speed.

$T = 1\frac{15}{60}$ hours

$T = 1\frac{1}{4}$ hour

$D = 60$Km

$S = D/T$

$S = 60 \div \frac{5}{4}$

$S = \frac{12}{60} \times \frac{4}{5}$

$S = 12 \times 4$

$S = 48$Km/hr

Section B (60 Marks)

21. **The list below shows prices of different items in a certain shop.**

2kg of wheat flour at sh.8000

1kg of sugar at sh.4600

4 bars of soap at sh.14000

A half litre of milk at sh.900.

(a) How much money will Mafabi pay for 5kg of wheat flour? *(02 marks)*

2Kg at Sh.8000

5Kg at Sh.$\left(\frac{5 \times 8000}{2}\right)$

$=$ Sh.20,000

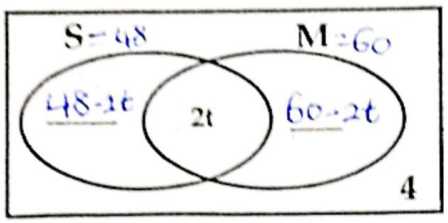
(b) Nafula buys **1kg** of sugar, **2 litres** of milk and **4 bars** of soap, how much does she pay? *(03 marks)*

Sugar	Milk	Soap	Total
Sh.4600	$\frac{1}{2}$L = Sh.900 2L = $(2 \times 900) \div \frac{1}{2}$ $= 1800 \times \frac{3}{1}$ $=$ Sh.3600	Sh.14000	Sh.14000 +Sh.4600 Sh.3600 Sh.22200

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22 In a class, 48 pupils like Science (S), 60 like Maths (M), 2t like both subjects while 4 do not like any of the two subjects.

(a) Complete the Venn diagram. *(02 marks)*



(b) If there are 100 pupils in the class, find the value of t *(03marks)*

$48-2t+2t+60-2t+4=100$

$48+60+4-2t=100$

$112-2t=100$

$112-112-2t=100-112$

$-2t=-12$

$\frac{-2t}{-2}=\frac{-12}{-2}$

$t=6$.

23 Omondi deposited some money in a bank that offers a simple interest rate of 18% p.a. After 12 months, Omondi had an amount of sh.885,000 in the bank.

(a) Calculate the amount of money Omondi deposited in the bank. *(03marks)*

$A=P+S.I$

$SI=A-P$ —(i)

$SI=P\times R\times T$ —(ii)

put (i) in (ii)

$A-P=P\times R\times T$

$885000-P=P\times\frac{18}{100}\times\frac{12}{12}$

$885000-P=P\times\frac{18}{100}\times 1$

$100\times(885000-P)=\frac{18}{100}P\times 100$

$18P=88500000-100P$

$18P+100P=88500000-100P+100P$

$\frac{118P}{118}=\frac{88500000}{118}$

$P=$ sh. 750,000

(b) Calculate Omondi's interest. *(02marks)*

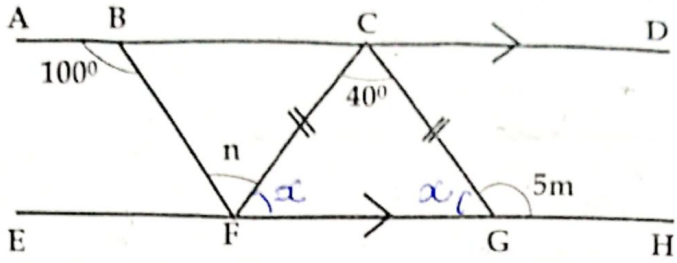
interest = Amount − principal

$=885000-750,000$

$=$ sh. 135000

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24 **In the figure below, AD is parallel to EH. Study the figure and answer the questions that follow.**



(a) Find the value of **m**. *(03 marks)*

$x + x + 40^\circ = 180^\circ$

$2x + 40^\circ = 180^\circ$

$2x + 40^\circ - 40^\circ = 180^\circ - 40^\circ$

$\frac{2x}{2} = \frac{140^\circ}{2}$

$x = 70^\circ$

$x + 5m = 180^\circ$

$70^\circ + 5m = 180^\circ$

$70^\circ - 70^\circ + 5m = 180^\circ - 70^\circ$

$\frac{5m}{5} = \frac{110^\circ}{5}$

$m = 22^\circ$

(b) Find the value of **n**. *(02 marks)*

$n + x = 100^\circ$

$n + 70^\circ = 100^\circ$

$n + 70^\circ - 70^\circ = 100^\circ - 70^\circ$

$n = 30^\circ$

$\therefore n = 30^\circ$

25 Namenya filled a cylindrical tank with water using **10 containers** of capacity **12.32 litres.** If the base area of the cylindrical tank was **616cm²**, find the height of the water in the cylindrical tank.

1 container $= 12.32\,L$

10 containers $= (10 \times 12.32)\,L$

$= 123.2\,L$

Volume $=$ capacity $\times 1000$

$= 123.2 \times 1000$

$= 123200\,cm^3$.

$V =$ base area $\times$ height

$\frac{123200}{616} = \frac{616 \times h}{616}$

$200 = h$

$h = 200\,cm$.

(05 marks)

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26 **Wanume went to the market with sh.70,000 and bought the items shown in the table below. He was given a discount of sh.500 and went back home with sh.2000.**

Complete the table.

Item	Unit cost	Total cost
2kg of meat	sh 14000	sh 28000
3 fish	sh 6000	sh 18000
$1\frac{1}{2}$ kg of chicken	sh 15000	sh 22500
		sh 68500

÷ ×

Sh. 14000 × 2 = Sh. 28000

Sh. 18000 / 3 = Sh. 6000

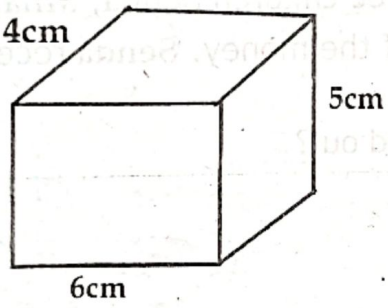
Total expenditure = 70,000 − (2000)
= 68000 + 500
= 68500/=

68500 − 46000 = 22500

22500 / 3/2
= 22500 ÷ 3/2
= 22500 × 2/3
= 15000

(05marks)

27 **Study the figure below and use it to answer the questions that follow.**



(a) Calculate its **T.S.A** (03marks)

T.S.A = 2(L×W) + 2(2(L×h) + 2(W×h))
= 2(6×4) + 2(6×5) + 2(4×5)
= 2(24) + 2(30) + 2(20)
= 2×24 + 2×30 + 2×20
= 48 + 60 + 40
= 148 cm^2

(b) Work out its volume. (02marks)

V = l×w×h
V = 6cm × 4cm × 5cm
V = 120 cm^3

28 A parent spends $\frac{2}{7}$ of her salary on food and the rest on fees. If she spends sh.210,000 more on fees than food. Calculate her total monthly salary. (05marks)

Fees $= \frac{7}{7} - \frac{3}{7}$
$= \frac{5}{7}$

more $= \frac{5}{7} - \frac{2}{7}$
$= \frac{3}{7}$

let her monthly salary be m.

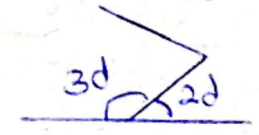
$\frac{3}{7} \times m = 210{,}000$

$\frac{1}{7} \times \frac{3}{7} m = 210{,}000 \times 7$

$\frac{1}{3} m = \frac{210{,}000 \times 7}{3_1}$

m = sh. 490,000

29 The interior and exterior angles of a regular polygon are 3d and 2d respectively. Name the polygon.



$3d + 2d = 180°$
$\frac{5d}{5} = \frac{180°}{5}$
$d = 36°$

ext $\angle$ = 2d
= 2 × 36°
= 72°

$n = \frac{360°}{\text{ext} \angle}$

$n = \frac{360°}{72°}$

n = 5 sides

∴ It is a pentagon.

(05 marks)

30 A mother shared part of her money to three children Senda, Mila, and Kilo. Mila received $\frac{1}{3}$ while Kilo received $\frac{1}{5}$ of the money. Senda received $\frac{1}{2}$ of the remainder.

(a) What fraction of the money was shared out? (03 marks)

Mila	Kilo	Remainder	Senda	Total
$\frac{1}{3}$	$\frac{1}{5}$	$\frac{15}{15} - \frac{8}{15}$	$= \frac{1}{2}$ of $\frac{7}{15}$	$= \frac{8}{15} + \frac{7}{30}$
$\frac{1}{3} + \frac{1}{5} = \frac{5+3}{15}$		$= \frac{7}{15}$	$= \frac{1}{2} \times \frac{7}{15}$	$= \frac{16+7}{30}$
$= \frac{8}{15}$			$= \frac{7}{30}$	$= \frac{23}{30}$

(b) If the mother remained with sh49,000, how much money did she have at first? (02marks)

$= \frac{30}{30} - \frac{23}{30}$
$= \frac{7}{30}$

let her total amount at first be n.

$\frac{7}{30} \times n = 49000$

$\frac{1}{30} \times \frac{7}{30} n = 49000 \times 30$

$\frac{1}{7} n = \frac{49000 \times 30}{7_1}$

$n = 7000 \times 30$

n = Sh. 210000

31 A man was 27 years older than his son 10 years ago. His wife is four years younger than he is. If their total age is 116 years now, how old is the son? (04marks)

Let the son's age be x.

	son	man	wife	Total
10 years ago	x	$x+27$		
now	$x+10$	$x+27+10$	$x+37-4$	116

$x+10+x+37+x+37-4=116$

$x+x+x+10+37+37-4=116$

$3x+84-4=116$

$3x+80-80=116-80$

$\frac{3x}{3}=\frac{36}{3}$

$x=12$

$\therefore$ Son $=12+10$

$=22$ years

(b) How old is the wife? (01marks)

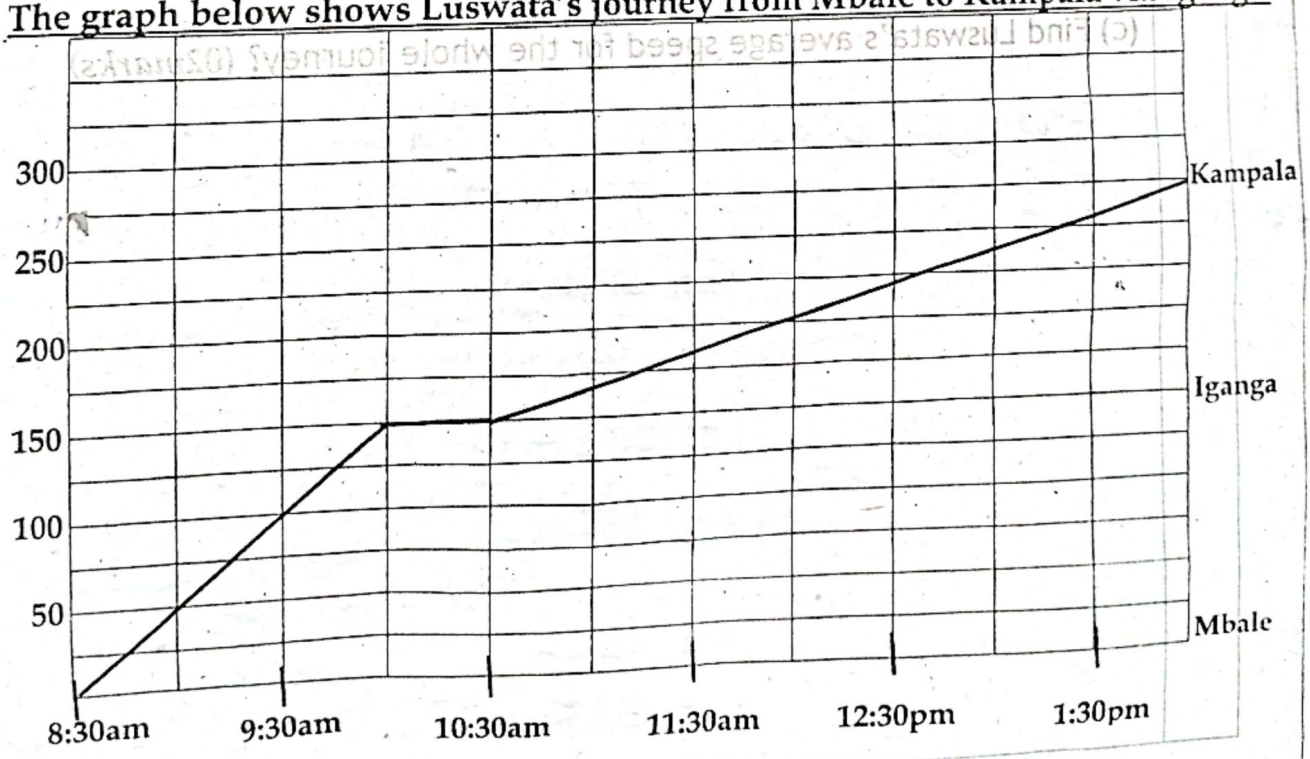
Wife $=x+33$

$=x+33$

$=12+33$

$=45$ years.

32 The graph below shows Luswata's journey from Mbale to Kampala via Iganga.



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P T O

(a) Where was Luswata at 10:15am? *(01marks)*

Iganga

(b) At what speed was Luswata travelling at from Mbale to Iganga? *(02marks)*

D = 150km

T = 1½ hours

$S = \frac{D}{S}$

$= 150 \div \frac{3}{2}$

$= \frac{5}{150} \times \frac{2}{3}$

$= 50 \times 2$

$= 100 km/hr$

(c) Find Luswata's average speed for the whole journey? *(02marks)*

$\text{Average Speed} = \frac{\text{total distance}}{\text{Total time}}$

$= \frac{275 km}{5\frac{1}{2} hours}$

$= 275 \div \frac{11}{2}$

$= \frac{25}{275} \times \frac{2}{11}$

$= 25 \times 2$

$= 50 km/hr$